DEUS Platform

Server Specification & Capacity Plan for Multi-Tenant Hosting

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This document specifies the server-side software requirements and capacity considerations for hosting the DEUS platform, a multi-tenant customer-relationship management application, at **deus.lucenai.eu**. It is intended to support the hosting provider in confirming whether existing infrastructure is sufficient or whether additional capacity is required, and to give internal stakeholders a consolidated view of the deployment plan.

Contents

1. Executive summary

The DEUS platform is a multi-tenant CRM application currently in the launch-readiness phase. Existing application code already supports logical multi-tenancy via organisation-scoped database rows. To meet the operational requirement of providing each client with a dedicated login at **deus.lucenai.eu/<client>** with a separate database under direct operator control, three deployment patterns are available — described in Section 2. The recommended pattern is **physical database-per-tenant**, which combines strong isolation with manageable operational overhead and a modest application-side change (approximately 150 lines of code for tenant resolution and connection-pool factory).

The recommended hosting target is a single dedicated server in an EU datacentre, GEX44-class or larger (Ryzen 7 8700GE-class CPU, 64 GB RAM, 1 TB NVMe). This configuration supports approximately 80 to 120 concurrent tenants under the recommended pattern. Scaling beyond this threshold requires either a larger server class (EX101 / 128 GB RAM and above) or a horizontal split across multiple servers, which is outside the scope of the initial launch.

All required server software is open-source or operates as a managed service. No commercial licences are required.

2. Tenancy architecture options

Three distinct deployment patterns address the multi-tenant requirement. Each implies a different per-tenant footprint and different operational characteristics. The choice is made once at the start of deployment and is not trivially reversible.

Option	Model	Per-tenant footprint	Capacity (64 GB)	Operational overhead
A	Logical multi-tenancy (shared database, row-level scoping)	Negligible (MB on disk)	500 – 1,000+	Low
B	Physical database-per-tenant (shared application process)	≈ 150 MB RAM ≈ 300 MB disk	80 – 120	Moderate
C	Full instance-per-tenant (dedicated process and DB)	≈ 500 MB RAM Dedicated DB	15 – 25	High

Capacity figures assume a steady-state workload of approximately 50 active users and 10,000 contacts per tenant.

Recommended pattern — Option B

Option B is recommended for the launch deployment. It provides each client with a dedicated MariaDB database under direct administrative control, while keeping a single application process per server. The implementation requires a thin tenant-resolution layer in the application — described in Section 6 — and carries the operational overhead of one Prisma migration step per tenant when schema changes are released. Option C should be reserved for individual client contracts that require process-level isolation (for example, a client subject to regulated-industry contractual requirements).

3. URL routing pattern

Two routing patterns are technically feasible for distinguishing tenants:

- **Path-based:** `deus.lucenai.eu/<client>` — a single domain with the tenant identifier in the path.
- **Subdomain-based:** `<client>.deus.lucenai.eu` — a separate subdomain per tenant under a wildcard certificate.

Aspect	Path-based	Subdomain-based
TLS certificate	Single certificate for the apex domain	Wildcard certificate via DNS-01 challenge
Browser cookie isolation	Cookies on the apex are visible to every tenant — explicit Path scoping required	Automatic — browser confines cookies to the issuing subd
Application code complexity	Application must be aware of the path prefix in every link and asset	Application is unaware of the prefix — minimal changes
DNS configuration	One A record	One A record plus a wildcard A record
Operational complexity	Higher	Lower

Recommended pattern — subdomain-per-tenant

The subdomain-based pattern is recommended. A wildcard A record at the registrar points all subdomains to the dedicated server; Caddy issues per-host certificates automatically via the DNS-01 challenge. Cookie isolation between tenants is enforced by the browser without additional application code. The path-based pattern remains technically achievable but requires explicit cookie-path scoping in the authentication layer and per-tenant configuration for the OAuth callback handlers.

4. Server software inventory

The following components must be installed on the host. All are open-source and available through standard package repositories or upstream-supplied installers. The combined fresh-install footprint is approximately 8 GB on disk.

Layer	Component	Version	Purpose
Operating system	Ubuntu LTS or Debian stable	24.04 LTS / 12	Host operating system
Runtime	Node.js	22 LTS	JavaScript application runtime
Runtime	pnpm	9	Package manager
Runtime	PM2	latest	Process supervision and log rotation
Web layer	Caddy	2	TLS termination and reverse proxy
Database	MariaDB	11.4 LTS	Per-tenant application data store
Database	PostgreSQL	17	Authentication store
Cache	Redis	7	Rate-limit token bucket and session cache
Email	Resend (managed service)	—	Transactional email — no server-side install
Backups	b2 CLI (Backblaze)	latest	Daily database snapshots to EU bucket
Security	ufw, fail2ban, OpenSSH, unattended-upgrades	—	Firewall, intrusion mitigation, key-only SSH, security patches

Observability	Sentry SDK (managed service)	—	Application error tracking — no server-side install
Observability	journalctl, logrotate	—	System log retention — included with the operating system
Build tools	git, build-essential	—	Source control and native module compilation

Components not required

- Container runtime (Docker, Podman) — the application runs directly under PM2.
- Container orchestrator (Kubernetes, Nomad) — single-host deployment.
- Additional reverse proxies (Apache, Nginx) — Caddy is the sole proxy layer.
- Standalone DNS server — DNS is managed at the registrar.
- Network load balancer — single-host deployment.
- Commercial database licences — both MariaDB and PostgreSQL are open-source.

5. Resource sizing and capacity envelopes

Resource consumption is decomposed into a constant baseline and a per-tenant increment. Figures below assume the recommended Option B (physical database-per-tenant) deployment with Caddy, Node.js, MariaDB, PostgreSQL and Redis active.

Per-tenant footprint

Resource	Cost per tenant
Disk — MariaDB database files	150 – 500 MB
Disk — PostgreSQL authentication row	Less than 1 MB
RAM — MariaDB connection pool (10 active connections)	80 – 100 MB
RAM — InnoDB buffer pool slice (shared)	30 – 50 MB
RAM — Application-side Prisma client	20 MB
Total per tenant (steady state)	≈ 150 MB RAM, ≈ 300 MB disk

Shared baseline (constant regardless of tenant count)

Resource	Cost
Operating system and systemd	1 GB RAM
Caddy 2	100 – 200 MB RAM
Node.js application under PM2	500 – 800 MB RAM
MariaDB server (excluding buffer pool)	1 GB RAM
PostgreSQL server	500 MB RAM
Redis server	256 – 512 MB RAM
MariaDB InnoDB buffer pool (tunable)	16 – 24 GB RAM
Backup staging area on disk	1 – 2 GB
Total baseline RAM	≈ 20 – 28 GB
Total baseline disk	≈ 10 GB

Capacity envelopes by server class

Server class	RAM	Tenants — Option B	Tenants — Option C
AX42-class	32 GB	30 – 50	5 – 10
GEX44-class (recommended)	64 GB	80 – 120	15 – 25
EX101-class	128 GB	250 – 400	50 – 80
AX162-class	256 GB	600 – 900	150 – 250

Disk is rarely the limiting resource at this scale: 1,000 tenants × 500 MB = 500 GB, well within a 1 TB NVMe drive.

Network

- 1 Gbps unmetered or 100 TB/month allowance is sufficient for a CRM workload.
- Typical bandwidth consumption is below 50 GB per tenant per month, dominated by API traffic and asset transfer.
- Inbound webhooks (payment provider, calendar providers) are negligible in volume.

6. Application-level changes for per-tenant database isolation

The current application code uses a single database per environment. Implementing Option B requires four discrete additions:

Component	Approximate size	Description
Tenant resolver middleware	≈ 50 lines	Reads the tenant slug from the request hostname or path, looks the te
Per-tenant Prisma client factory	≈ 80 lines	Returns a Prisma client bound to the tenant's database connection stri
Tenant registry table	1 PostgreSQL table	Single source of truth listing each tenant — slug, display name, MariaD
Provisioning script	≈ 50 lines	Creates the MariaDB database and user, runs the application's Prisma

Total estimated effort is approximately 150 lines of TypeScript, plus the SQL for the registry table and migration scaffolding for new tenants. No changes to existing application logic, business rules, or user interface are required. The change is additive and does not affect tenants already deployed under the logical-tenancy model.

Connection-pool sizing

Each tenant's Prisma client is configured with an initial connection limit of 5. At 100 tenants × 5 connections = 500, this matches the recommended `max_connections = 500` setting on MariaDB. Should the tenant count exceed 100, either the global maximum can be raised or the per-tenant limit can be reduced to 3.

7. Cutover sequence

The cutover from the existing single-tenant environment to the multi-tenant configuration follows the steps below. Each step is a single operator action; the entire sequence can be completed in a single working day once the hosting provider has confirmed the specification.

Step	Action	Owner
1	Hosting provider confirms the specification and provisions the dedicated server.	Hosting provider
2	Bootstrap script applied — installs operating-system packages, databases, web layer, monitoring agents.	Operator
3	Wildcard A record (*.deus.lucenai.eu) configured at the registrar; root A record points to th	Operator
4	Caddy is configured with on-demand TLS and the DNS-01 challenge plugin for the registrar's DNS provider.	Operator
5	Tenant resolver middleware and Prisma factory deployed; provisioning script smoke-tested against a sandbox tenant.	Operator
6	First production tenant provisioned end-to-end and validated.	Operator
7	Second tenant onboarded as the first commercial customer.	Operator

From step 7 onward, every subsequent tenant is provisioned by a single shell command and is fully operational within approximately five minutes.

8. Information requested from the hosting provider

To enable a complete capacity assessment and to confirm the suitability of existing infrastructure, the hosting provider is requested to confirm the following items:

1. Server hardware specification — CPU model and core count, RAM capacity, NVMe storage capacity.
2. Network configuration — dedicated link bandwidth and any monthly transfer cap.
3. Datacentre region — must be EU-resident for compliance with EU data-protection requirements (Falkenstein, Helsinki, Amsterdam and equivalent are all suitable).
4. Storage redundancy — RAID-1 or hardware-mirrored NVMe is required for production workloads; storage class for the boot volume.
5. Snapshot policy — frequency, retention period, and storage location for system snapshots.
6. Reverse-DNS / PTR record configuration — must be settable to `deus.lucenai.eu`.
7. IPv6 inclusion — preferred but not required.
8. Out-of-band access — KVM, IPMI or rescue-system access in case primary SSH is unavailable.
9. DDoS protection — confirmation of any network-layer protection included with the package.
10. Bandwidth-overage policy — pricing or throttling behaviour beyond the included allowance.

9. Recommendations summary

If existing dedicated server is GEX44-class (64 GB RAM) or larger

- **Existing capacity is sufficient** for the recommended Option B deployment up to approximately 80 – 120 tenants.
- No additional server is required for the initial launch.
- An additional server should be considered when the active-tenant count exceeds 100, to leave operational headroom.

If existing dedicated server is AX42-class (32 GB RAM) or smaller

- Existing capacity is workable for a soft-launch of 30 – 50 tenants under Option B.
- A second dedicated server should be planned at approximately tenant 25 to maintain operational headroom.
- Migration to a 64 GB-class server is recommended before onboarding customer 5 if Option C is contemplated for any client.

If procuring a new dedicated server

- **Floor specification: 64 GB RAM, 1 TB NVMe, EU datacentre.**
- Recommended specification: GEX44-class (Ryzen 7 8700GE, 64 GB RAM, 1 TB NVMe) or equivalent.
- Disk capacity is not the constraint at this scale.
- EU residency of the datacentre is mandatory for compliance with EU data-protection law.

10. Out of scope

The following items are intentionally not addressed by this document:

- **Application-level rate limiting.** Rate limits per tenant are enforced inside the application using Redis-backed token buckets.
- **Migration of existing logical-tenant data into per-tenant databases.** Addressed in a separate operational document and only relevant if the platform is currently serving production tenants under the logical model.
- **Per-tenant payment-provider configuration.** Each tenant maps to a separate subscription record at the payment provider; no infrastructure-side changes are required.
- **Email deliverability.** Handled by the managed email provider through DNS records (SPF, DKIM, DMARC) on the application domain. No server-side email infrastructure is provisioned.
- **Cross-region disaster recovery.** The initial launch is a single-region deployment with daily off-site backups to an EU object-storage bucket. Multi-region active-active replication is a future architectural step beyond the scope of this launch.

End of document — DEUS Platform Server Specification, version 1.0.